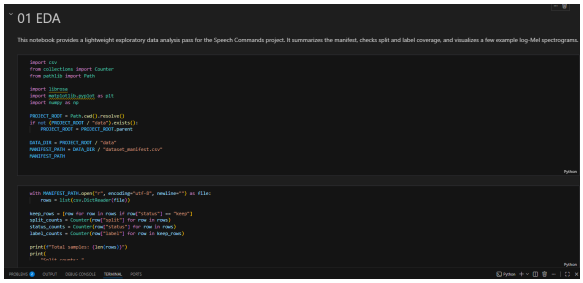
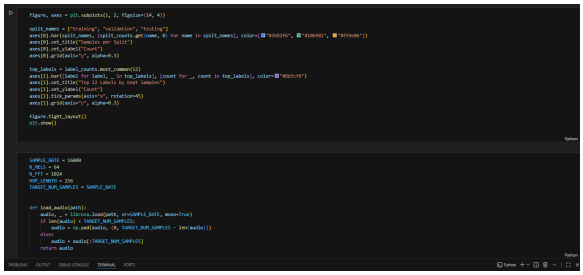
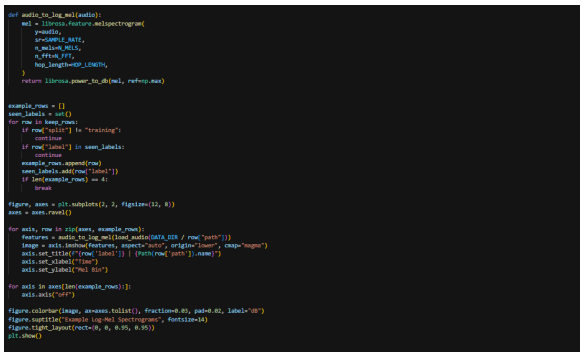
Data acquired/cleaned; splits finalized; EDA notebook.

The dataset acquired from arXiv titled Google Speech Commands was meticulously checked and cleaned for any audio files that might be corrupted, inaudible, or silent to prevent any mishaps when training the model.





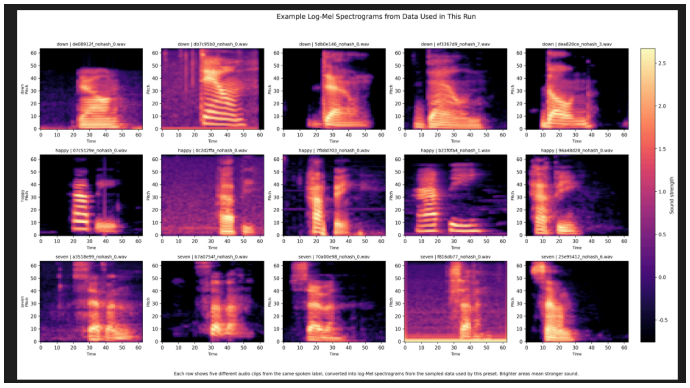


Baselines trained (simple ML + DL); initial metrics logged.

The history of initial logs recorded when training the model was saved in low, medium, and high-end. The screenshot below shows the logs for medium and high-end training.

CNN experiment running (first results); NLP component scaffolded/prototyped.

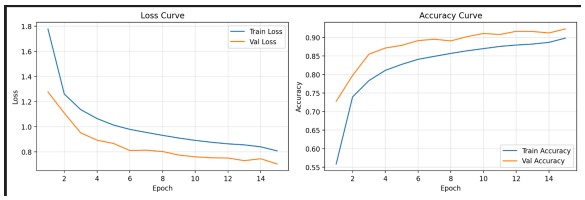
The data_pipeline.py will get all the available wav files and will then be moved to the train.py, which will use the audio files to train and detect audio and present them in the spectrogram below.



RL agent stubbed with reward design; early learning curves (even if noisy).

The initial reinforcement learning of the model below shows the early learning curve

and steadily improves as more training is experienced. And checks if any patterns align with the spectrogram.



Draft Model Card sections started; Ethics statement updated.

ETHICS STATEMENT

The Speech Command Recognition system carries three key ethical risks. First, accent and dialect bias arise because the Google Speech Commands v0.02 dataset skews toward American English speakers, potentially reducing accuracy for Filipino, British, or Indian accents; this is mitigated through audio augmentation and per-class performance documentation. Second, audio privacy risk exists since voice data is biometrically sensitive. MIT will be used for educational purposes and not for deployment to the public. collecting no personal audio and storing no raw audio in the repository. Third, command misinterpretation, where commands like "no" may be classified as "go" is mitigated by RL-based threshold tuning and fallback handling for low-confidence predictions. No PII (Personal Identifiable Information) is present in the dataset; all speakers are fully anonymized with random IDs. This system is an academic prototype only and is only intended for external use.

MODEL CARD

Field	Details
Model	CNN (Mel-spectrogram) + RL Threshold Agent
Dataset	Google Speech Commands v0.02 (Warden, 2018)
License	Creative Commons BY 4.0 license.
Size	105,829 clips, 35 classes, ~2,600 speakers
Splits	Provided validation/testing lists; medium_end used 10,000 / 1,000 / 1,000
Cleaning	252 invalid clips removed (<0.3% of total)
Input	1-sec audio → Mel-spectrogram (128 bins, 16kHz)
Targets	Accuracy ≥90%, Macro-F1 ≥0.88, RL cost reduction ≥20%
Limitation	People with strong accents; offline prototype only
Intended Use	Academic research is not only for deployment